



SUREN NETWORK
PLAY WITH TECHNOLOGIES

SNCNA

SUREN NETWORK CERTIFIED NETWORK ASSOCIATE

PING ERROR TEST

Explain proper scenario for following ping errors ?

Destination host unreachable

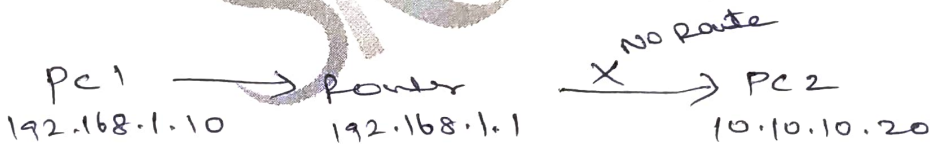
- Your PC is 192.168.1.10
 - You are to Reach 192.168.1.20
 - Your PC knows the destination is on the local n/w
 - But Cannot Reach the destination at layer 2
- Reason: Destination PC Power off, Cable problem, Switch/Port problem, wrong VLAN
Request timed out

- Your PC sends the ICMP Echo Request
- The Packet may leave your PC Successfully
- But no ICMP reply comes back within the time out period

Reason: Destination PC is down
Firewall is blocking
Router/Firewall drops the packet

Reply from some ip destination host unreachable

- You are trying to Reach 10.10.10.20
- But the message Comes from 192.168.1.1
- 192.168.1.1 usually you router/default gateway



TTL expired (Time to live)

- Every IP Packet has a TTL value
- Each Router decreases the TTL by 1 when forwarding packet
- If there is a routing loop, the packet keeps travelling between routers

(TTL 5) → R1 - TTL 4 → when TTL reaches 0, the router drops the packet's and sends an ICMP Time Exceeded message back to the source
 R2 - TTL 3
 R3 - TTL 2
 R1 - TTL 1
 R2 - TTL 0